

Software Developer Role - Task

Technical Assignment — Submission Form Below

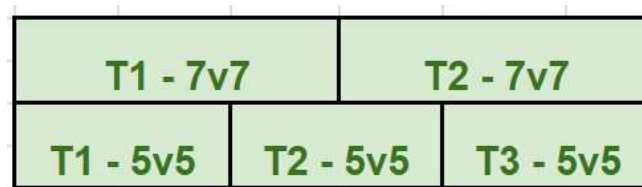
Submission deadline: 11 PM, 07 June, 2026

About Sportomic:

Sportomic is a sports venue management platform working with turfs, academies, and sports complexes across India. This task tests one thing: can you model a real-world physical constraint in clean, correct code — and reason about the edge cases before you write a line.

The Problem — Parent-Child Arena Booking

A turf can be split two ways. The same physical ground can be sold as 2 large courts (7v7) OR as 3 smaller courts (5v5). They share the same space — so booking one court must block every other court that physically overlaps it.



The Rules to Handle

- If one 5v5 is booked → one 7v7 gets blocked. 2 other 5v5 still available, 1 7v7 still available.
- If one 7v7 is booked → any 2 5v5 get blocked. 1 5v5 still available, 1 7v7 still available.
- If two 5v5 are booked → both 7v7 get blocked. 1 5v5 still available.
- If two 7v7 are booked → all 5v5 get blocked. No arena available.

Your Task

Build the availability engine. Given any set of current bookings, your code must return exactly which courts are still bookable and which are blocked — correctly, for every possible combination, not just the four above.

What to Submit

- **Working Code with UI** — any language. Runnable, with instructions.
- **Your Logic** — how you modelled the overlap, and why this design.
- **Test Cases** — the four rules above, plus any edge cases you caught.
- **Explainer video** — screen recording with voiceover

Deliverables

Item	Format
Code with UI	GitHub / Colab / Drive link
Writeup (Algorithm + Logic)	Google Doc / PDF link
Explainer video (screen recording with voiceover)	Link (Drive/YouTube)

Submit your task here: <https://forms.gle/tHLCgL55rNmbQ7E8>